

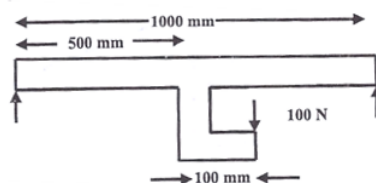
Chapter 2

Bending Moment and Shear Force Diagram

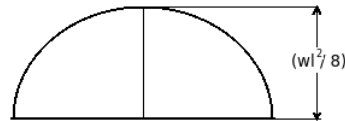
2.1 Question Bank UNIT - 2

Quiz Questions

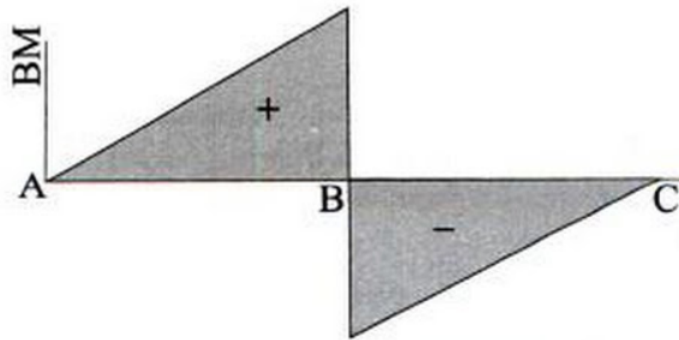
1. A simply supported beam of span length 6m and 75mm diameter carries a UDL of 1.5 kN/m. What is the maximum value of bending moment? (6.75kN-M)
2. A sudden jump anywhere on the bending moment diagram of a beam is caused by a?(*Moment*)
3. In a cantilever subjected to a combination of concentrated load, UDL, UVL, the maximum bending moment will occur at(*fixed end*)
4. In a simply supported beam carrying a uniformly distributed load over the left span indicates that the point of contra-flexure will occur at(*no point of contra-flexure*)
5. When the bending moment is parabolic curve between two points it indicates that there is a load (*UDL*)
6. What is the maximum bending moment for simply supported beam carrying a point load "W" kN at its centre?($WL/4$)
7. In a simply-supported beam loaded as shown below, the maximum bending moment is Nm is



8. Uniformly varying load between two sections in shear force diagram is represented by
9. In bending moment diagram, if no load acts between two sections, then it is represented by.....
10. The diagram shown below, depicts bending diagram for

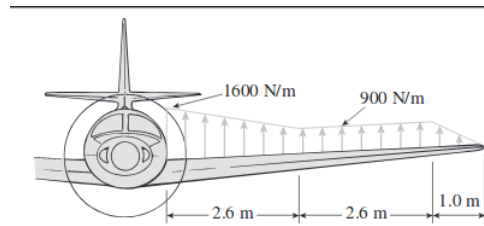


11. If the bending moment of a simply supported beam is as shown, the the load acting on the beam is

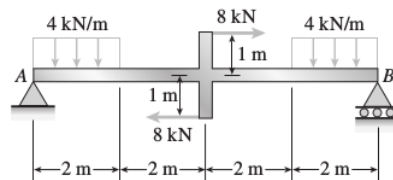


12. A simply supported beam is subjected to a couple somewhere in the span, it would produce.....
13. A cantilever with span L and Load F at the free end. The Bending Moment at the support will be
14. For the shear force to be uniform along the beam it should carry a anywhere in its span (*moment/couple*).
15. If a simply supported beam at its end is loaded by a central concentrated load then maximum moment is M . If the same weight is equally distributed over the beam then the maximum bending moment will be.....($M/2$)
16. The rate of change of Bending moment is equal to the at that section (*shear force*)
17. Point of contra-flexure is where changes sign
18. At the point of contra-flexure the shear force ins the shear force diagram will be
19. Bending moment variation due to concentrated loads will be(*Linear*)
20. The bending moment at a point on a beam is the algebraic sum of all the on either side of the point.(*moment*)

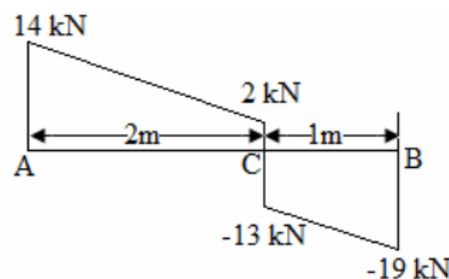
Main Questions



1. Under cruising conditions the distributed load acting on the wing of a small air plane has the idealized variation shown in the figure. Calculate the shear force V and bending moment M at the inboard end of the wing.
2. A simple beam AB is loaded by two segments of uniform load and two horizontal forces acting at the ends of a vertical arm (see figure). Draw the shear-force and bending-moment diagrams for this beam.

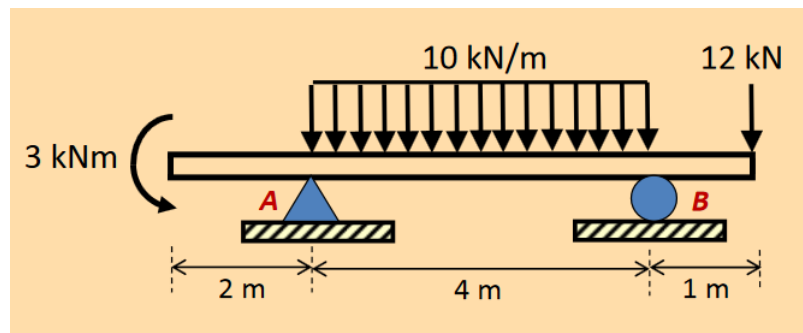


3. Derive the relationship between load, shear force and bending moment.
4. A Cantilever beam carrying gradually varying load from zero at fixed end and w /unit length at the free end. Draw the shear force and bending moment diagram.
5. Prove that the maximum bending moment is $wL^2/12$ for a simply supported beam of length l is subjected to a symmetrical uniformly varying load with zero intensity at the ends and intensity w (load per unit length) at the mid span.
6. Draw the loading conditions and the bending moment diagram for the given shear force diagram. Determine the boundary conditions of the beam.

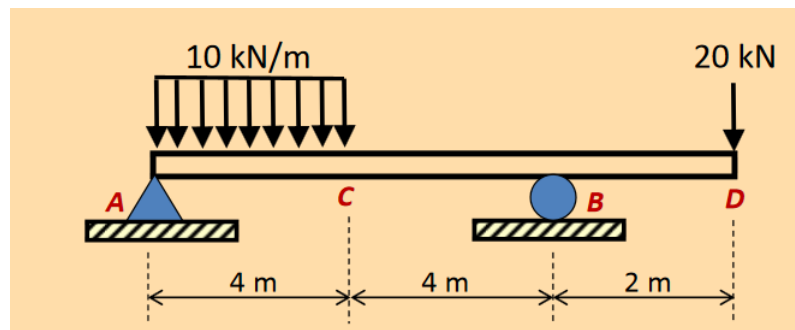


7. A Cantilever beam carrying gradually varying load from zero at fixed end and w /unit length at the free end. Draw the shear force and bending moment diagram. Also derive the relationship between load, shear force and bending moment. What can you conclude?

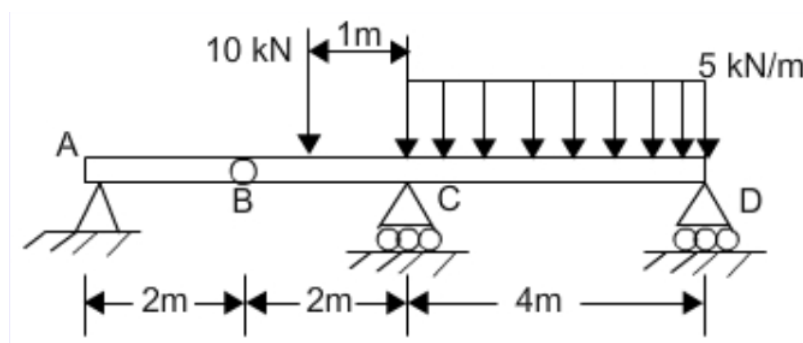
8. Prove that when a beam is subjected to a constant bending moment along its length, then the shear force will be zero at all sections along the beam.
9. A beam AB is hinged-supported at its ends and is loaded by couple $P \cdot c$ at the center of the beam. Determine the magnitude or shearing force at a section x . Also draw the shear force bending moment diagram for the same.
10. Prove that the maximum bending moment is $0.0645wl^2$ for a simply supported beam of length L subjected to a UDL Wx/L .
11. Calculate the shear force and bending moment for the beam(s) subjected to the loads as shown in the figure(s) then draw the shear force diagram (SFD) and bending moment diagram (BMD).



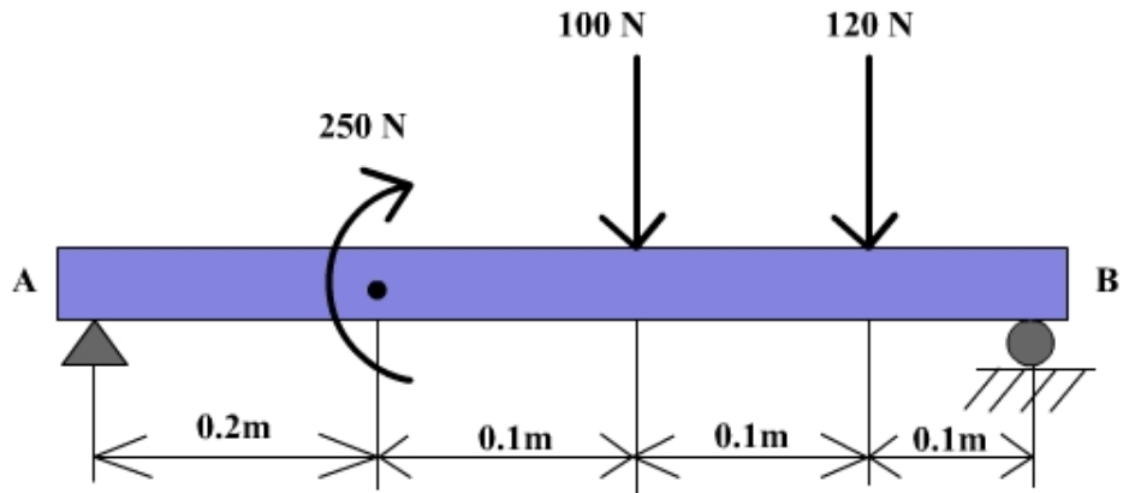
(a)



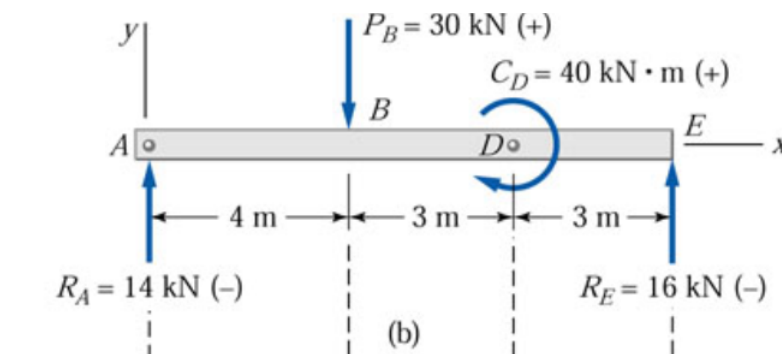
(b)



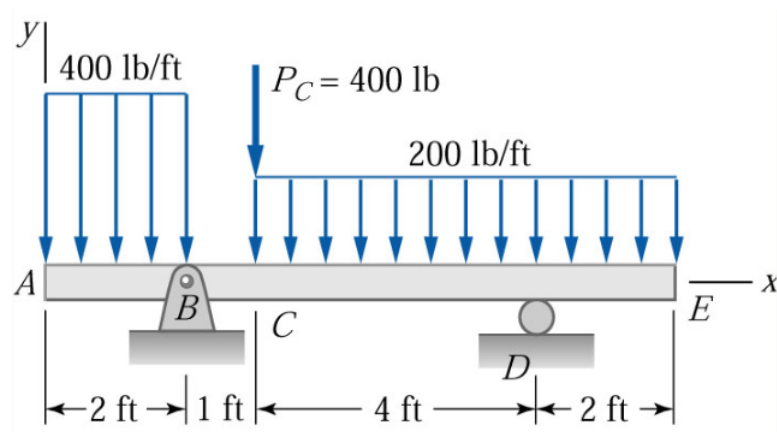
(c)



(d)



(e)



(f)